



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

ISE 333: User Interface Development (Design)

Chapter 11: Communication and Collaboration

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Introduction

- Social Nature

- Constant and immediate communication and interaction with family, friends, collaborators, colleagues, coworkers, and even pets are now commonplace in the increasingly networked world

- Representative websites

- Twitter
- Facebook
- Weibo



Introduction

- Attention on Negative Aspects
 - It is also important for UIDers to consider and account for the downsides and negative exigencies of communication and collaboration UIs
 - Criminals, terrorist recruiters, and oppressive political leaders can use social network sites for negative purposes
 - A miscreant can bully or deceive others online, and hate groups can spew their propaganda
 - People may inadvertently share photos or other information that damages their reputation and is difficult to remove from their online profile
 - Social media can make behaviour such as stalking and public shaming easier
 - Ideas can become polarized when liberal and conservative thinkers cluster together
 - UIDers must be aware of such behaviours and possible outcomes and consider UI options that circumvent or mitigate the worst possibilities
 - Good UID, effective community leadership, and thoughtful governance policies and strategies can lead to more positive social outcomes

Introduction

- Various Modalities of Online Communication
 - Social network sites
 - Despite their huge popularity, they are just one particular form of online communication
 - Text messaging
 - Voice calls
 - Video calls
 - E-mail
 - Online conference
 - Online teaching
- Satisfying Various Human Needs
 - Different communication channels and tools are more or less suited for different tasks and human needs

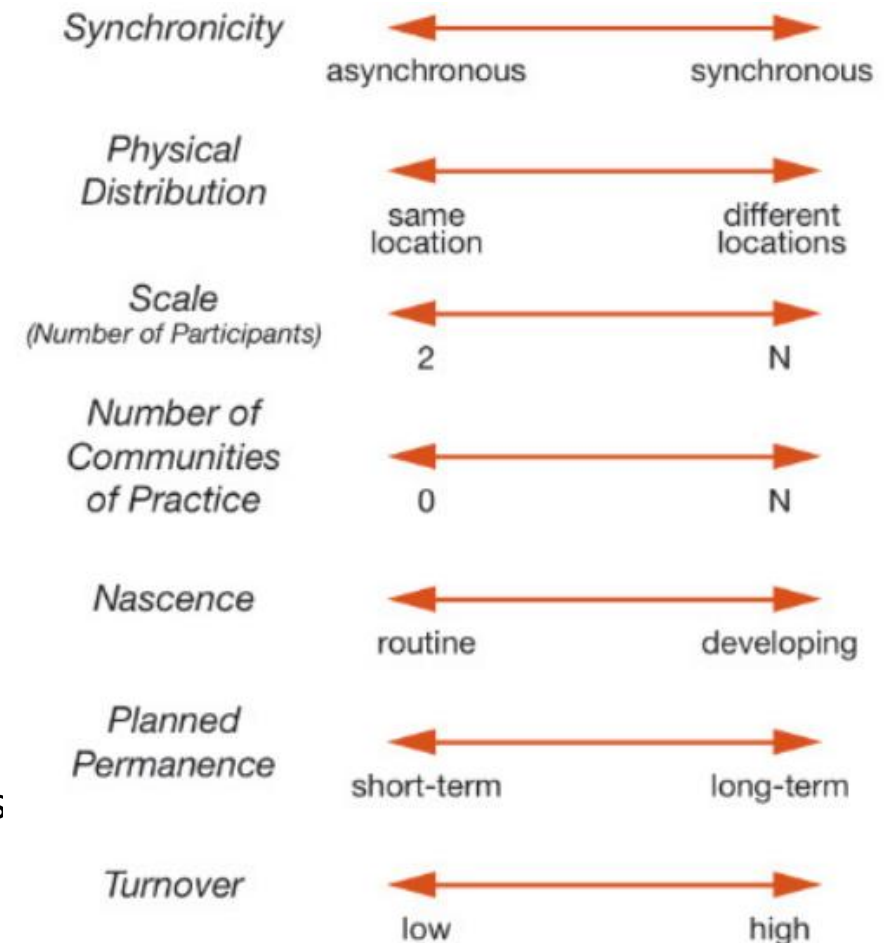


Models of Collaboration

- Time/Space Matrix
 - The traditional way to decompose collaborative UIs, which has four quadrants
 - Same time (synchronous), same place (co-located)
 - Shared table display, wall display
 - Same time (synchronous), different place (remote)
 - Teleconferencing
 - Different time (asynchronous), same place (co-located)
 - Public display
 - Different time (asynchronous), different place (remote)
 - E-mail, discussion forums, version control

Models of Collaboration

- More Dimensions
 - Universal UID for collaboration and communication systems fluidly accommodates users across the spectrum of the following dimensions
 - Synchronicity
 - Physical distribution
 - Scale of the collaboration
 - Number of communities of practice
 - Nascence
 - Time of existence of collaboration
 - Planned permanence
 - Turnover
 - Refers to the stability of the people involved in a collaboration in terms of how frequently new participants enter and leave the group



Specific Goals and Contexts

- Conception
 - People collaborate because doing so is satisfying or productive
 - Collaboration allows individuals to reap the emotional rewards of socializing and interacting with others, to accomplish greater goals than they could alone, or to meet and transact with people who they otherwise could not
- Communication and Conversation
 - One of the essential coordinated actions that most people participate in on a daily basis is exchanging ideas, information, and knowledge with other people via conversation
- Meeting Coordination
 - Sometimes users merely need the aid of communication tools to coordinate a real-world meeting time and location such that groups and communities can come together as needed to coordinate IRL (in real life)

Specific Goals and Contexts

- Online Markets
 - Buying, selling, and trading have been driving commerce for a few thousand years, but the past twenty years have seen the nature of these activities change dramatically as people adopt the internet for shopping, trading, buying, selling, and delivery of goods
 - To cope with the scale of people, items, and content that is present in online markets, collaborative filtering algorithms have been developed
 - One approach to collaborative filtering works by representing the purchase or preference data of each individual in the market and then identifying other people who have a similar profile

Specific Goals and Contexts

- Creative Production and Crowdwork
 - Big creative projects demand that users work together
 - Work needs to be broken down into pieces and re-assembled, contingencies and interdependencies require planning, quality must be ensured, different roles and skills must be brought to bear, and supporting administrative duties underlie collaboration
 - Various platforms and tools exist to facilitate these activities
 - Social coding platforms like GitHub
 - File-sharing tools like Dropbox
- Entertainment and Gaming
- Education
 - Interactive courses available online from platforms like Coursera and Udacity have come to be known as Massive Open Online Courses (MOOCs)

UID Considerations

- Cognitive Factors
- Individual Factors
- Collective Factors

UID Considerations

- Cognitive Factors
 - Common ground
 - Establishing common ground (i.e. the knowledge that communicators have in common) as well as jointly understood references during communication can be essential for effective collaboration
 - Social cues
 - There are a variety of other nonverbal cues that can also enhance communication, including facial expressions, gaze direction, posture, proximity, and bodily orientation
 - Activity awareness
 - Making social behaviour visible facilitates awareness and ultimately accountability for one's actions

UID Considerations

- Individual Factors
 - Privacy
 - On one hand, if activity is collected implicitly based on actions within the system (rather than explicitly recorded or set by the user), this could affect the use or adoption of the system, as users may not want others to see every little action they take
 - On the other hand, in many cases, it is important to know who contributed what in a collaborative work project so that issues, corrections, or applause can be directed and so that the result can be better understood
 - Identity
 - Online communities open up questions regarding how people represent and portray themselves when people's physical bodies are not shown
 - Text or avatars become the primary medium of communication
 - Trust and reputation
 - Motivation

UID Considerations

- Individual Factors
 - Privacy
 - Identity
 - Trust and reputation
 - Related to identity is the notion of reputation and the ability to develop a sense of trust around that reputation
 - Motivation
 - There is a strong intrinsic motivation for interpersonal connectedness
 - There are a wealth of other reasons for people's collaboration and communication
 - Altruism
 - Reciprocity
 - Reputation or status
 - Habit

UID Considerations

- Collective Factors
 - Deviance
 - People are not always perfect, and in some cases they may not be aware of the specific norms of the community they are participating in
 - UIDers must be keen to consider various ways in which deviant behaviour can be regulated or to make social norms more apparent and salient so that non-normative behaviours are reduced and their impacts on the community lessened
 - Moderation
 - One of the approaches to cope with deviant behaviour is to have moderators evaluate contributions and take various actions on the postings
 - Policies and norms
 - They can be important signals to users in online communities so that users know what constitutes acceptable versus unacceptable behaviour and so that protocols for moderation or other decisions are apparent
 - Policy documents are often posted in places where users can easily find them